EXP4:

Design a simple Topology and configure one Router, two Switches and PCs using cisco Packet Tracer

Aim:

To design and configure a simple network topology consisting of one router, two switches and four pcs in cisco packet tracer, assign IP addresses, configure router interfaces and verify communication between different LANs using the ping command.

Build the Topology:

→ Add 1 Router, 2 cisco 2960 switches and 4 PCs

→ Connect:

- → PC0 and PC1 → Switch 0
- → PC2 and PC3 → Switch 1
- Switch 0 → Router Gig0/0/0
- Switch 1 → Router Gig0/0/1

→ Use copper straight-through cables for the connections

Configure PC IP addresses.

Assign the following static IP addresses:

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	192.168.10.11	255.255.255.0	192.168.10.1
PC2	192.168.20.10	255.255.255.0	192.168.20.1
PC3	192.168.20.11	255.255.255.0	192.168.20.1

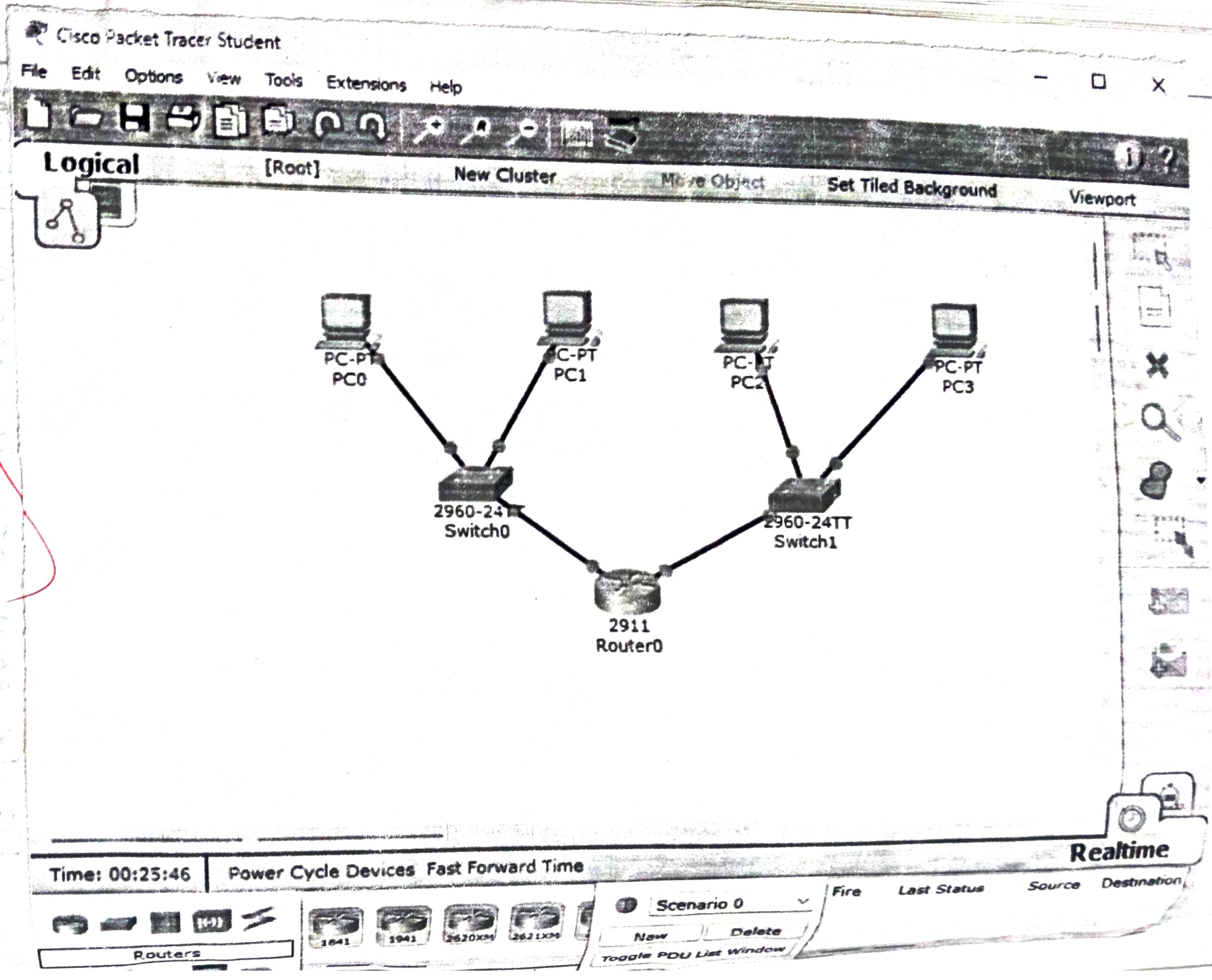
Configure the Router

In the CLI, enter:

enable

configure terminal

interface g0/0/0


Cisco Packet Tracer Student
File Edit Options View Tools Extensions Help
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
PC-PT
PC0
C-PT
PC1
PC-PT
PC2
PC-PT
PC3
2960-24
Switch0
2960-24TT
Switch1
2911
Router0
Time: 00:25:46
Power Cycle Devices
Fast Forward Time
Realtime
Routers
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination

ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface g0/0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
end
write memory

Verify connectivity
From PC0, open command prompt & test

ping 192.168.10.1
ping 192.168.20.10

Result:
The topology is successfully configured PCs within both LANS can communicate through the router. The first ping may time out due to ARP resolutions, but subsequent pings should receive successful replies.